

Yifan Han

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EDUCATION AND RESEARCH APPOINTMENT

Institute of Automation, Chinese Academy of Sciences (CASIA) <i>M.Res. Candidate, National Laboratory of Pattern Recognition; GPA: 3.85/4.00; Supervisor: Prof. Wenzhao Lian</i>	Beijing, China 2023 – 2025
Shanghai Jiao Tong University (SJTU) <i>Research Assistant, School of Artificial Intelligence; Supervisors: Prof. Wenzhao Lian</i>	Shanghai, China 2025 – Present
Rutgers University (RU) <i>Research Intern, Department of Computer Science; Supervisor: Prof. Jingjin Yu</i>	New Brunswick, NJ, USA 2026 – Present
Huazhong University of Science and Technology (HUST) <i>B.Eng., Qiming Class, School of Artificial Intelligence and Automation; GPA: 3.78/4.00</i>	Wuhan, China 2020 – 2024

RESEARCH EXPERIENCE

Research Assistant, School of Artificial Intelligence, SJTU <i>Under the supervision of Prof. Wenzhao Lian and Prof. Yao Mu</i>	2025 – Present Shanghai, China
– Conduct research in two primary directions: dexterous manipulation, including dexterous-hand policy learning and RL-based post-training; and affordance learning. – Contributed to active perception, VLA-based robotic systems, and MLLM-based grounding modules for manipulation tasks.	
Research Intern, Algorithmic Robotics & Control Lab <i>Department of Computer Science, Rutgers University; Supervisor: Prof. Jingjin Yu</i>	2026 – Present New Brunswick, NJ, USA
– Worked on task-conditioned scene-level affordance grounding for real-time robotic manipulation in complex multi-object scenes. – Contributed to affordance-conditioned policy learning.	
Research Student, NLPR / Institute of Automation, CASIA <i>Under the supervision of Prof. Wenzhao Lian</i>	2023 – 2025 Beijing, China
– Worked on learning-based robot manipulation, including multi-task imitation learning, long-horizon manipulation, and 3D articulated-object understanding. – Contributed to projects on human-to-robot skill transfer, affordance-centric manipulation, and active robotic perception.	

PUBLICATIONS

First-Author Publications

- [1] BORA: Bridging Offline Token-Action RL and Online Residual Adaptation for Real-World Dexterous VLA Models.**
Zhongxi Chen*, Yifan Han^{*,†}, Yanming Shao, Huanming Liu, Congsheng Xu, Yao Mu, Wenzhao Lian.
*Under Review. *Equal contribution; †Project leader.*
Brief Description: Bridges offline token-action reinforcement learning with online residual adaptation for robust real-world dexterous VLA deployment.
- [2] BridgeACT: Bridging Human Demonstrations to Robot Actions via Unified Tool-Target Affordances.**
Yifan Han[†], Jianxiang Liu, Haoyu Zhang, Yuqi Gu, Hanqing Wang, Zhongxi Chen, Litao Liu, Wenzhao Lian.
Under Review. †Project leader.
Brief Description: Introduces role-conditioned tool-target affordances as an embodiment-agnostic representation for transferring manipulation knowledge from human videos to robots.
- [3] Affordance2Action: A Benchmark and Framework for Task-Conditioned Part Affordance Grounding in Real-Time Robot Manipulation.**
Litao Liu*, Yifan Han^{*,†}, Pengfei Yi, Wenbo Yu, Hanqing Wang, Haoran Du, Enze Yuan, Zilin Yuan, Ruiding Feng, Michael Liu, Qi Zhang, Jingjin Yu.
*Under Review. *Equal contribution; †Project leader.*
Brief Description: Builds a benchmark and real-time framework for task-conditioned part affordance grounding in robot manipulation.
- [4] DexHiL: A Human-in-the-Loop Framework for Vision-Language-Action Model Post-Training in Dexterous Manipulation.**
Yifan Han*, Zhongxi Chen*, Yuxuan Zhao, Congsheng Xu, Yanming Shao, Yichuan Peng, Yao Mu, Wenzhao Lian.
*Under Review. *Equal contribution.*
Brief Description: Builds a human-in-the-loop post-training framework for dexterous-manipulation VLA models, combining human feedback and policy optimization.

- [5] **FSAG: Enhancing Human-to-Dexterous-Hand Finger-Specific Affordance Grounding via Diffusion Models.**
Yifan Han, Yichuan Peng, Pengfei Yi, Junyan Li, Hanqing Wang, Gaojing Zhang, Qi Peng Liu, Wenzhao Lian.
Under Review.
Brief Description: Studies finger-specific affordance grounding from human hands to dexterous hands using diffusion models.

SELECTED PUBLICATIONS AS MAIN CONTRIBUTOR

- [1] **FoAM: Foresight-Augmented Multi-Task Imitation Policy for Robotic Manipulation.**
Litao Liu, Wentao Wang, Yifan Han, Zhuoli Xie, Pengfei Yi, Junyan Li, Yi Qin, Wenzhao Lian.
AAAI 2026, Oral Presentation.
Brief Description: Contributed to foresight-augmented multi-task imitation learning, introducing task-progress prediction and visual representations to improve generalization in robotic manipulation.
- [2] **Splatter Joint: 3D Gaussian Splatting for Articulated Objects.**
Junyan Li, Yifan Han, Pengfei Yi, Wenzhao Lian.
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2025.
Brief Description: Contributed to 3D Gaussian Splatting for articulated-object modeling, focusing on movable-part representation, structural understanding, and view-consistent reconstruction.
- [3] **SAGE: Scene Graph-Aware Guidance and Execution for Long-Horizon Manipulation Tasks.**
Jialiang Li, Wenzheng Wu, Gaojing Zhang, Yifan Han, Wenzhao Lian.
IEEE International Conference on Robotics and Automation (ICRA), 2026.
Brief Description: Contributed to a scene-graph-guided long-horizon manipulation framework that integrates object-relation modeling with execution planning for complex multi-stage tasks.
- [4] **Affordance-R1: Reinforcement Learning for Generalizable Affordance Reasoning in Multimodal Large Language Models.**
Hanqing Wang, Shaoyang Wang, Yiming Zhong, Zemin Yang, Jiamin Wang, Zhiqing Cui, Jiahao Yuan, Yifan Han, Mingyu Liu, Yuexin Ma.
AAAI 2026, Oral Presentation.
Brief Description: Contributed to a reinforcement-learning framework for affordance reasoning in MLLMs, improving generalizable operation-region understanding across objects and scenes.
- [5] **Look, Zoom, Understand: The Robotic Eyeball for Embodied Perception.**
Jiashu Yang, Yifan Han, Yucheng Xie, Ning Guo, Wenzhao Lian.
Preprint / Manuscript, 2025.
Brief Description: Contributed to an active embodied perception system that improves robot understanding of task-relevant regions and object details through viewpoint adjustment and local observation.
- [6] **Viewpoint Matters: Dynamically Optimizing Viewpoints with Masked Autoencoder for Visual Manipulation.**
Pengfei Yi, Yifan Han, Junyan Li, Litao Liu, Wenzhao Lian.
Preprint / Manuscript, 2026.
Brief Description: Contributed to MAE-based dynamic viewpoint optimization, analyzing and improving the robustness of visual manipulation policies under changing viewpoints.

HONORS AND AWARDS

National First Prize and Third Place Overall , The 21st National Robotics Competition for College Students (ROBOCON)	2022
National First Prize , Robot Equestrian Event, The 21st National Robotics Competition for College Students (ROBOCON)	2022
National First Prize , The 22nd National Robotics Competition for College Students (ROBOCON)	2023
National Third Prize , Robot Equestrian Event, The 22nd National Robotics Competition for College Students (ROBOCON)	2023
National Second Prize , The 5th China University Intelligent Robot Creativity Competition	2022
Self-Improvement Scholarship , Huazhong University of Science and Technology	2022

ACADEMIC SERVICE

Reviewer for IROS, ICRA, IEEE Robotics and Automation Letters (RA-L), AAAI, CVPR, and ECCV.

SKILLS

Programming: Proficient in Python, MATLAB, C, LaTeX; Advanced in C++, Java Script

Software: MuJoCo, Isaac Sim, SolidWorks, Adobe (Photoshop), Microsoft Office (Certificate of Level II)

Hardware: Robot: Franka, Xarm Robots, Universal-Robots; Other: Micro Controller Unit (51 MCU, STM32), RealSense, Orbbec Camera